

CRISPR-Edited DPSCs, Constitutively Expressing BDNF Enhance Dentin Regeneration in Injured Teeth

Reviewed Preprint

v1 • March 25, 2025

Not revised

Ji Hyun Kim, Muhammad Irfan, Sreelekshmi Sreekumar, Stephanie Kim, Atsawasuwan Phimon, Seung Chung 

Department of Oral Biology, College of Dentistry, University of Illinois Chicago, Chicago, United States •

Department of Orthodontics, College of Dentistry, University of Illinois Chicago, Chicago, United States

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This study provides a potentially **useful** investigation into the positive role of BDNF/TrkB signaling in implanted dental pulp stem cells to enhance dentin regeneration in the context of dental caries. Some of the key methods used need to be much better documented, and the data should be strengthened and added to in support of several of the claims of functional benefit, which are **inadequately** supported at present. Additional details for the validation of the reagents and techniques are needed to support the interpretation of the results.

<https://doi.org/10.7554/eLife.105153.1.sa3>

Abstract

Dental caries is one of the most common health issues worldwide arising from the complex interactions of bacteria. In response to harmful stimuli, desirable outcome for the tooth is the formation of tertiary dentin, a protective reparative process that generates new hard tissue. This reparative dentinogenesis is associated with significant inflammation, which triggers the recruitment and differentiation of dental pulp stem cells (DPSCs). Previously, we have shown that brain-derived neurotrophic factor (BDNF) and its receptor TrkB, key mediators of neural functions, are activated during the DPSC-mediated dentin regeneration process. In this study, we further define the role of inflammation in this process and apply stem cell engineering to enhance dentin regeneration in injured teeth. Our data show that TrkB expression and activation in DPSCs rapidly increase during odontogenic differentiation, further amplified by inflammatory inducers and mediators such as TNF α , LTA, and LPS. An in vivo dentin formation assessment was conducted using a mouse pulp-capping/carries model, where CRISPR-engineered DPSCs overexpressing BDNF were transplanted into inflamed pulp tissue. This transplantation significantly enhanced dentin regeneration in injured teeth. To further explore potential downstream pathways, we conducted transcriptomic profiling of TNF α -treated DPSCs, both with and without TrkB antagonist CTX-B. The results revealed significant changes in gene expression related to immune response, cytokine signaling, and extracellular matrix interactions. Taken together, our study advances our understanding of the role of

BDNF in dental tissue engineering using DPSCs and identifies potential therapeutic avenues for improving dental tissue repair and regeneration strategies.

Introduction

Dental caries, commonly known as tooth decay, remains one of the most prevalent chronic diseases, affecting a significant portion of the global population (Peres et al., 2019 [↗](#)). This condition results from the complex interplay between bacterial biofilms, dietary sugars, and host factors, leading to the demineralization of tooth enamel and dentin (Bowen et al., 2018 [↗](#); Mosaddad et al., 2019 [↗](#)). In recent years, there has been growing interest in the role of dental pulp stem cells (DPSCs) in regenerative dentistry due to their ability to differentiate into dentin-forming odontoblast-like cells and their potential in tooth repair and regeneration (Potdar & Jethmalani, 2015 [↗](#)). It is widely recognized that the regeneration of the dental-pulp complex is closely associated with inflammation, especially since dental caries and the subsequent regenerative processes take place within an inflammatory environment (Galler et al., 2021 [↗](#)). Inflammation serves as a vital biological response for ensuring host survival in the face of infection and tissue injury, playing a crucial role in maintaining normal tissue homeostasis (Peiseler & Kubes, 2018 [↗](#)). It is generally accepted that the regeneration of the injured tooth is closely tied to inflammation (Lin & Rosenberg, 2011 [↗](#)). Specifically, it plays a pivotal role in the regeneration of injured dental tissues by stimulating the recruitment, proliferation, and differentiation of pulp progenitor cells and odontoblasts (Nakashima et al., 2013 [↗](#)). However, an imbalance between inflammation and repair can lead to irreversible tissue damage. While inflammation is recognized as necessary for proficient tissue regeneration, only a few studies have explored its role in dentin regeneration.

The brain-derived neurotrophic factor (BDNF) and its receptor, tropomyosin receptor kinase B (TrkB), constitute a critical signaling pathway involved in the survival, development, and function of neurons (F. Zhang et al., 2012 [↗](#)). Emerging evidence suggests that the BDNF-TrkB pathway also plays a significant role in the inflammatory response and tissue repair processes (Cappoli et al., 2020 [↗](#); Hang et al., 2021 [↗](#)). The overexpression of BDNF in DPSCs has been shown to enhance their regenerative capabilities, which may be particularly beneficial in the context of dental caries where inflammation and tissue damage are prominent. Indeed, our previous study demonstrated the critical role of BDNF/TrkB in the DPSC odontoblastic differentiation under inflammation. (Kim, Irfan, Hossain, George, et al., 2023) TNF α (Tumor Necrosis Factor- α)-stimulated BDNF/TrkB showed an enhanced odontoblast differentiation of DPSCs.

Although the general principles for successful BDNF-induced regeneration have been proposed, achieving clinical success remains challenging. One significant obstacle is the clinical application of BDNF itself, as the recombinant protein has an extremely short half-life, which greatly diminishes its effectiveness (Miranda-Lourenço et al., 2020 [↗](#)). Therefore, establishing a stable and continuous BDNF production platform is essential, and stem cell engineering could potentially address this critical need in the future. Considering the essential role of BDNF/TrkB signaling activation in the inflammatory dentinogenesis observed in cell culture conditions, there is a strong case for validating BDNF/TrkB data in vivo. In this study, we further explore the role of various inflammatory agents in modulating two BDNF receptors, TrkA and TrkB, as well as the potential therapeutic applications of BDNF overexpression in DPSCs using CRISPR technology for the treatment of dental caries. By uncovering the mechanisms through which the BDNF-TrkB pathway affects DPSC function and inflammation, we aim to offer insights into innovative regenerative strategies that could enhance dental health outcomes.

Result

Various inflammatory mediators increase TrkB expression and activation in DPSCs

Our previous study demonstrated that TNF α augments the odontoblast differentiation of DPSCs in a TrkB-dependent manner (Irfan & Chung, 2023 [DOI](#); Kim, Irfan, Hossain, George, et al., 2023). The literature indicates that pro-inflammatory cytokines, such as TNF α , LTA (lymphotoxin-alpha), and LPS (lipopolysaccharides), play significant roles in the formation of tertiary dentin (Irfan, Kim, Marzban, et al., 2022). To further confirm the role of inflammation in this process, various inflammatory mediators were applied during DPSC odontoblastic differentiation. The outline of the experimental timeline is shown (Figure 1A [DOI](#)). Human DPSCs were seeded on Day 0, followed by culture in normal growth media. On Day 3, the cells were transferred to dentinogenic media. TNF α , LTA, and LPS treatments were initiated every other three days. The representative image shows the immunofluorescence staining results for TrkB and phosphorylated TrkB (p-TrkB) under various treatments (Figure 1B [DOI](#)). The quantification of fluorescence intensity across these different treatment groups was analyzed by percentages of control (Figure 1C [DOI](#)). TrkB expression and activation in DPSCs were significantly upregulated during odontogenic differentiation, especially under inflammatory stimulants such as TNF α , LTA, and LPS by 301 ± 17 , 320 ± 15.2 , and 250 ± 19 , respectively vs control 165 ± 12.4 ($p < 0.01$). The spatial distribution of fluorescence intensity for TrkB and p-TrkB was measured across different treatment groups (Figure 1D [DOI](#)). Figures were analyzed using ZEN (blue) software, and graphs were created. This spatial analysis confirms the enhanced and localized expression of TrkB and p-TrkB in response to inflammatory stimuli.

Next, we examined the expression levels of TrkA and TrkB receptors in response to various inflammatory stimuli in normal and dentinogenic media. The results indicate significant differential expression of these receptors under both conditions. TrkB expression was modulated by the media and stimuli (Figure 2C [DOI](#)). Under control conditions, no significant differences were observed between the media types. However, LPS, LTA, TNF α , and C5a significantly increased TrkB expression in dentinogenic media compared to normal media ($p < 0.001$ for LPS, TNF α , and C5a; $p < 0.05$ for LTA). IL-5 caused a slight but significant increase in TrkB expression in dentinogenic media ($p < 0.05$). Conversely, baseline TrkA expression showed no significant difference between normal and dentinogenic media (Figure 2A [DOI](#)), and C5a and IL-5 did not induce significant changes in TrkA expression in either medium.

Figures 2B [DOI](#) and 2D [DOI](#) show the effect of TNF α stimulation on the expression of TrkA and TrkB and their phosphorylation vs control. 2D and 2.5 models show fluorescence spikes to evident the phenomenon. It is obvious that TNF α stimulation caused significant phosphorylation of TrkB ($p < 0.01$) compared to TrkA ($p < 0.05$). Graphs (Fig. 2B [DOI](#) 1 and D1) show significant changes among TNF α treated cells vs control. So, we chose TrkB as a target receptor for our further study.

The dose-dependent response of TrkB to TNF α (Figure 2E [DOI](#)) revealed minimal baseline expression at 0 ng/mL TNF α , with significant increases at 0.1 ng/mL, 1 ng/mL, and a pronounced peak at 10 ng/mL in dentinogenic media ($p < 0.001$). At 20 ng/mL TNF α , TrkB expression remained significantly higher in dentinogenic media compared to normal media ($p < 0.001$). The time-dependent response to a fixed TNF α concentration (10 ng/mL) showed that TrkB expression in dentinogenic media was significantly higher on day 1, further increased by day 3, and peaked on day 7 ($p < 0.001$) (Figure 2F [DOI](#)). Although TrkB levels slightly decreased on days 10 and 14, they remained significantly elevated compared to normal media. These findings emphasize the importance of the local microenvironment in influencing receptor expression and inflammatory responses.

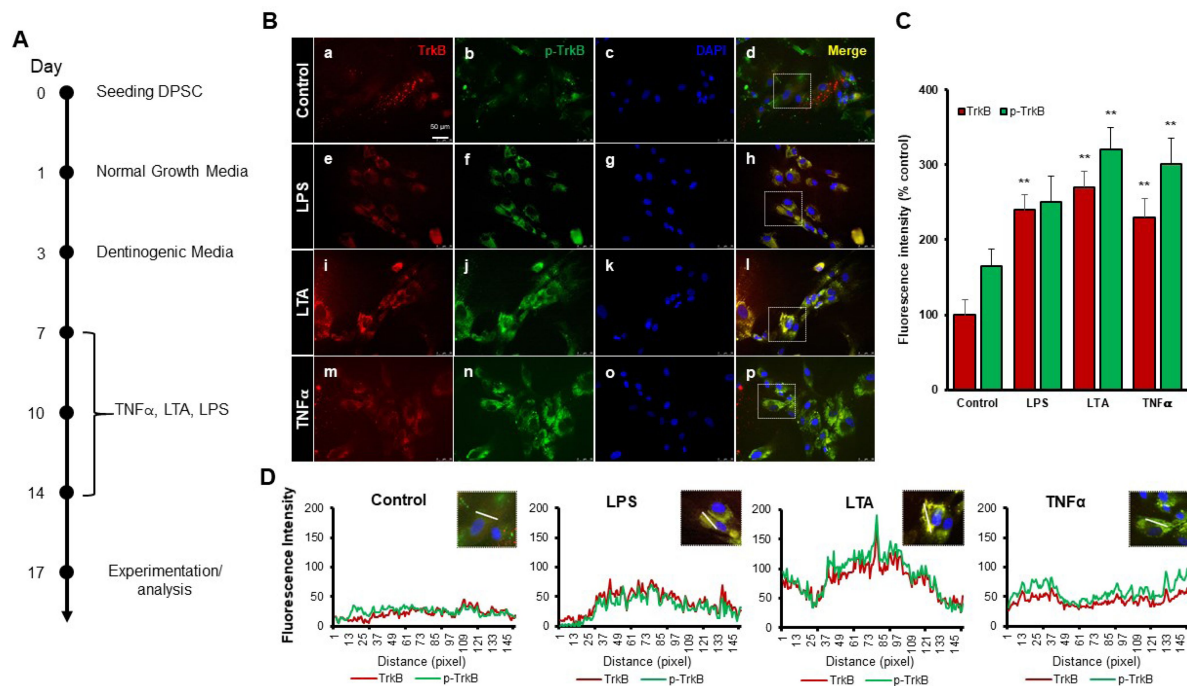


Figure 1.

Effects of various inflammatory mediators on the expression of TrkB and its phosphorylation in human DPSCs.

(A) Schematic representation of DPSC differentiation stimulated by various inflammatory mediators. **(B)** DPSCs were cultured in normal growth media with or without LPS, LTA or TNF α for 24 hours, and cells were fixed, and double immunofluorescent staining was performed (TrkB: red; p-TrkB: green). Cells were counter-stained with DAPI (blue). (a-d) the control cells showed less TrkB and p-TrkB expression than the LPS (e-h), LTA (i-l), or TNF α (m-p) treated groups. **(C)** The bar graph shows a significant increment in TrkB and p-TrkB expression in the groups treated with inflammatory mediators. * $p < 0.05$ and ** $p < 0.01$ vs control. **(D)** The Line graph shows that the co-localization of TrkB and p-TrkB are higher peaks observed in LPS, LTA, and TNF α treated groups than in control. Scale bar: 50 μ m.

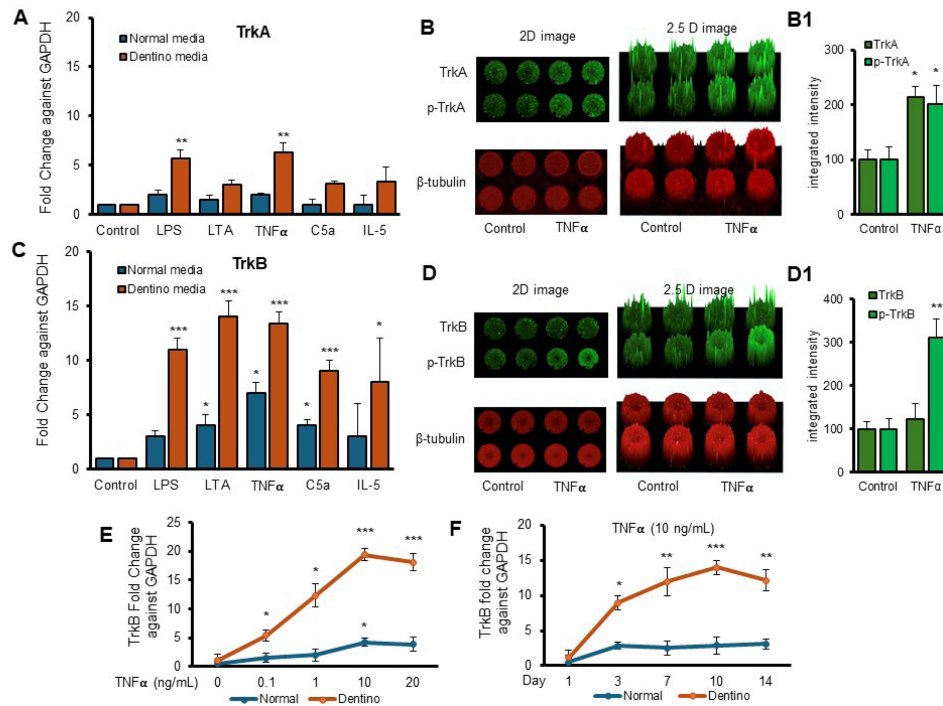


Figure 2.

Time-course analyses of TrkB mRNA expression in DPSCs.

(A and C) Real-time PCR was used to determine the effect of various inflammatory mediators (LPS, LTA, IL5, TNF α , and C5a) on the expression level of TrkA & TrkB mRNA in DPSCs after 1 d treatment in regular growth medium (blue bars) or dentinogenic medium (red bars). While these inflammatory components do not affect much the expression of TrkA or TrkB in normal growth medium, inflammatory mediators such as LPS, LTA, and TNF α stimulation increase TrkB expression in DPSCs undergoing odontoblastic differentiation (orange bars “LPS, LTA & TNF α ” vs. orange bar “Control”). Comparatively, TrkA expression changed little during the odontoblastic differentiation of DPSCs (A). **(B and D)** In cell western assay showing effects of TNF α on the expression of TrkA & TrkB and their phosphorylation in 2D and 2.5 D models. Graphs (B1 and D1) show fluorescence intensity TNF α vs control. **(E)** Effect of TNF α on TrkB expression. DPSCs were cultured in regular or dentinogenic/osteogenic medium and with 0, 0.1, 1, 10, or 20 ng/mL of TNF α . TrkB mRNA expression was determined by real-time PCR analysis after 24 h of culture. TNF α stimulation, independent of the concentration used, quickly potentiates the odontogenesis-regulated expression of TrkB. However, this TrkB expression increase is sustained with increasing concentrations of TNF α treatment, even at 20 ng/mL. **(F)** Real-time PCR was used to evaluate the level of TrkB mRNA in DPSCs after 1, 3, 7, 10, and d 14 of culture in regular medium (blue lines) or dentinogenic medium (orange lines) in TNF α -stimulated DPSCs odontoblastic differentiation. In contrast to relatively stable TrkB levels in undifferentiated DPSCs, TrkB expression quickly increased in dentinogenic medium, with the most significant increase detected between 7 and 10 d (fold increase: 7 d = 12.04 ± 2.1 , 10 d = 14.1 ± 1.72). Results are expressed as relative expression to glyceraldehyde 3-phosphate dehydrogenase (GAPDH). Data are presented as mean $\pm$ SD of 3 independent experiments. * P < 0.05 vs. regular medium, * P < 0.05, and ** p < 0.01, *** p < 0.001 vs. dentinogenic medium and without inflammatory stimulation.

TrkB expression is increased in human carious teeth and in DPSCs

Regenerative odontogenic differentiation in the context of caries occurs within an inflammatory environment, making it essential to understand the impact of inflammation on DPSCs to comprehend dentin regeneration fully (Kim, Irfan, Hossain, Shin, et al., 2023).

Immunofluorescence double staining was employed to analyze the expression of STRO-1, which is a stem cell marker, and TrkB in human carious and normal teeth, as well as in DPSCs (**Fig. 3A-K**). The in vivo analysis revealed that STRO-1 was primarily localized around blood vessels, while TrkB was more abundantly expressed in the inflamed pulp tissue of carious teeth compared to normal teeth. Quantitative analysis revealed significantly higher TrkB expression in carious teeth ($p < 0.001$). In vitro, both STRO-1 and TrkB were expressed in control DPSCs, but TNF α treatment markedly increased TrkB expression (**Fig. 3L-T**). The data showed that TrkB expression was significantly elevated in TNF α -treated DPSCs ($p < 0.01$). These findings suggest that TrkB is upregulated in response to inflammation in both carious teeth and DPSCs.

Transplantation of GFP-tagged CRISPR-engineered BDNF-overexpressing DPSCs enhances dentin regeneration in a pulp-capping mouse model

Our previous data demonstrated that inflammation and BDNF-induced DPSC odontogenic differentiation increased the expression of odontogenic differentiation markers. (Kim, Irfan, Hossain, George, et al., 2023). To investigate whether BDNF-TrkB signaling controls dentin regeneration following inflammation in vivo, we utilized CRISPR-engineered BDNF-overexpressing DPSCs in a pulp-capping mouse model (**Figure 4A**). We initially utilized the Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) technique to activate BDNF expression in DPSCs. Transfection of DPSCs with the BDNF CRISPR activation plasmid was confirmed through Western blot analysis, which revealed increased BDNF expression compared to the control group (**Figure 4B**). Quantification of the integrated intensity relative to beta-actin further indicated a significant increase in BDNF expression in CRISPR-activated DPSCs (**Figure 4C**). Histological analysis of H&E-stained sections was performed to evaluate tissue morphology and surgery site from the transplantation of CRISPR-engineered BDNF-overexpressing DPSCs in a pulp-capping mouse model (**figure 4D**).

Following confirmation of successful CRISPR activation, we conducted the transplantation of CRISPR-engineered BDNF-overexpressing DPSCs on dentin regeneration in a pulp-capping mouse model. Micro-computed tomography (micro-CT) imaging was utilized to assess dentin volume in both sham control and experimental groups. The sham control group [**Figure 5A** (a-h)] exhibited a clear demarcation of the injured dentin area, highlighted by the white arrow indicating the drilling operation and the dotted line specifying the damaged region. In contrast, the experimental group treated with CRISPR-engineered BDNF-overexpressing DPSCs [**Figure 5A** (i-p)] showed a notable increase in dentin formation at the injury site. A quantitative analysis of dentin density (**Figure 5D**) revealed a significant increase in the dentin volume in the group treated with CRISPR-engineered BDNF-overexpressing DPSCs compared to the sham control group. Specifically, the dentin volume in the treated group (**Figure 5C**) was 1241 ± 51 mg HA/ccm, significantly higher than the control group (**Figure 5B**), which measured at 1169 ± 9 mg HA/ccm ($p < 0.05$).

To evaluate morphological changes after transplantation, H&E staining was employed to assess cellular and structural changes in the dentin and pulp tissue following the transplantation of CRISPR-engineered BDNF-overexpressing DPSCs in the pulp-capping mouse model. **Figure 5** displays representative images of the sham control group and surgery group. The overall morphology of the tooth structure (**Figure 4D**) and higher magnifications of the injured area highlighted by the box are provided (**Figure 5**). In contrast to the control group, the BDNF-DPSC

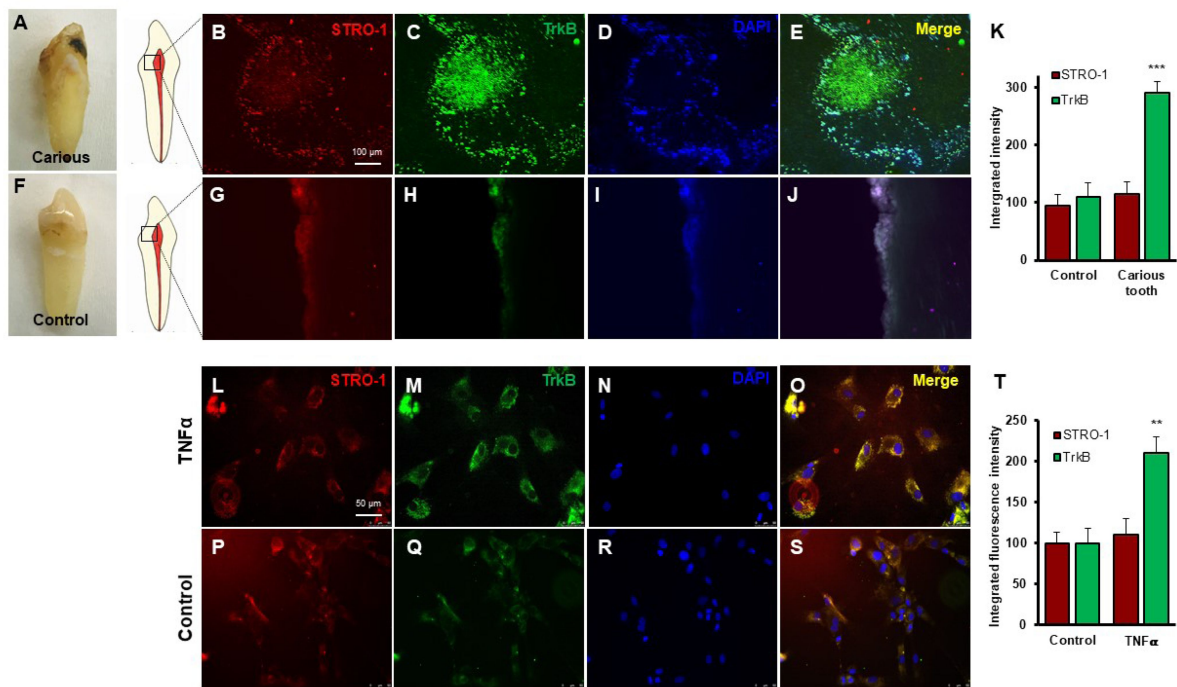


Figure 3.

Expression of STRO-1 and TrkB in human carious or normal tooth in vivo and DPSCs in vitro.

(A-J) Immunofluorescence double staining was used to localize STRO-1 and TrkB-expressing cells in the human virgin (control) and carious tooth section. While the expression of STRO-1 was detected in the vicinity of blood vessels, TrkB is expressed more in inflamed pulp tissue compared with virgin tooth. Merged images of STRO-1 and TrkB staining demonstrated a co-expression of TrkB and STRO-1 in the perivascular area. Scale bar: 100 μ m. Nuclei were counterstained with DAPI (blue). **(K)** The bar graph shows a significantly higher expression of TrkB in the carious tooth section (** $p < 0.001$ vs. control). **(L-S)** Expression of STRO-1 and TrkB by DPSCs in vitro. Immunofluorescence double staining was used to analyze TrkB expression in control or untreated and TNF α treated cells. All control DPSCs express both the mesenchymal stem cell marker STRO1 and TrkB, while TNF α treated cells TrkB expression was significantly higher than control. Merged images of STRO1 and TrkB revealed that all STRO-1-positive cells expressed TrkB. Nuclei were counterstained with DAPI (blue). Scale bar = 50 μ m. **(T)** The bar graph shows significantly higher expression of TrkB in TNF α treated cells ($p < 0.01$ vs. control).

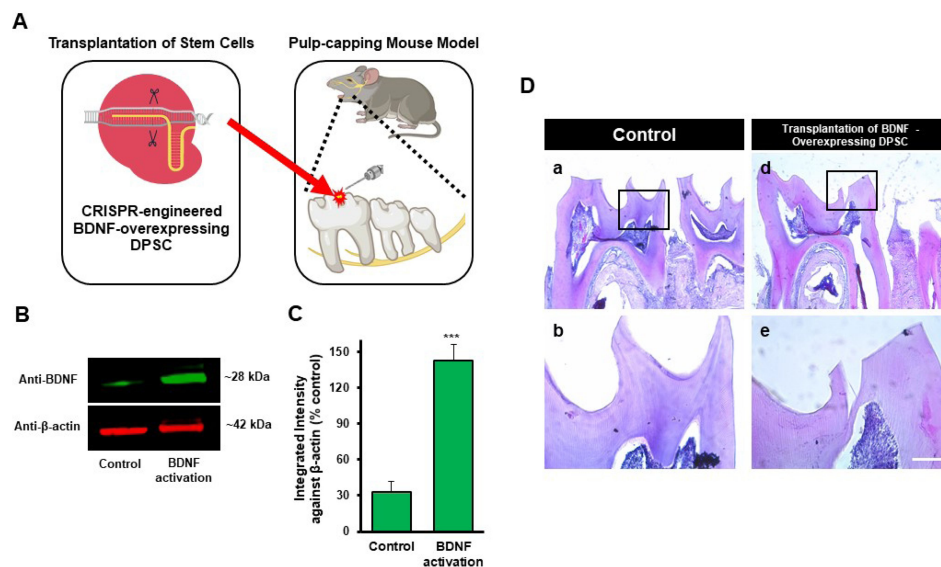


Figure 4.

The transplantation of CRISPR-engineered BDNF-overexpressing DPSCs in the pulp-capping mouse model.

(A) A schematic representation of the transplantation of DPSC into the first molar tooth after drilling. (B) The confirmation of BDNF CRISPR activation plasmid enhanced the expression of BDNF. (C) Bar graph showing the integrated intensity of CRISPR-engineered BDNF-activated DPSCs against β-actin compared to control. (D) H&E staining of sham control and injured tooth in mouse transplanted with CRISPR-engineered BDNF-overexpressing DPSCs.

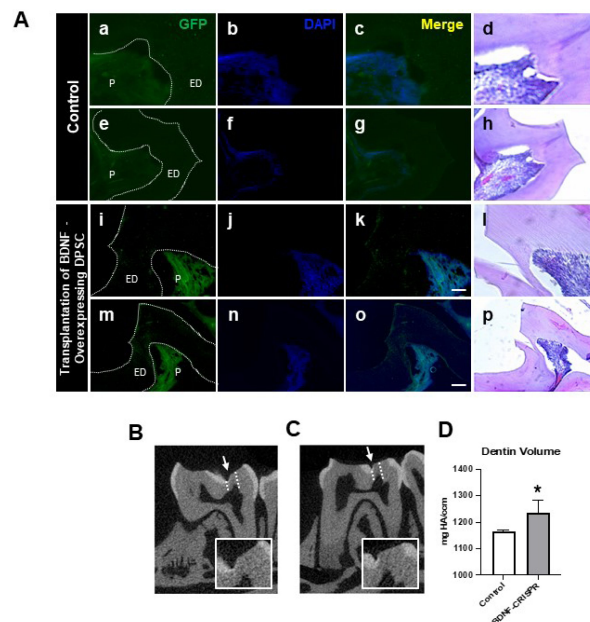


Figure 5.

The transplantation of CRISPR-engineered BDNF-overexpressing DPSCs in the pulp-capping mouse model and micro-CT analysis.

(A) Immunohistochemistry was performed to assess GFP-tagged transplanted cells. The white arrow indicates the pulp lining. (B) Micro-computed tomography (micro-CT) image in the Sham control of the pulp-capping mouse model. The white arrow indicates the drilling operation, and the dotted line specifies the injured area of dentin. The white box is a magnified image of the injured area. (C) The transplantation of CRISPR-engineered BDNF-overexpressing DPSCs in the pulp-capping mouse model. (D) Analyzed density of dentin compared with Sham control vs. transplantation of CRISPR-engineered BDNF-overexpressing DPSCs in the pulp-capping mouse model. ($p < 0.01$ vs. control).

transplantation group demonstrated significant dentin regeneration and repair. The newly formed dentin at the injury site appears more organized and closely resembles the native dentin structure.

GFP staining allowed us to identify and localize the transfected areas within the injury site (**Figure 5** [5A](#)–**6** [6A](#)). Specifically, GFP expression was observed in areas correlating with the injury compared to the control (**Figure 6A** [6A](#)). This confirmed the successful transfection and localization of the CRISPR-engineered DPSCs in the mandible of the mouse tooth. Line graphs show colocalizations of GFP-tagged DPSCs expressing BDNF and TrkB (**Figure 6B** [6B](#)). These results indicate that DPSCs overexpressing BDNF, engineered using CRISPR and transplanted into a pulp-capping mouse model, significantly improve dentin regeneration.

RNA sequencing of TNF α -induced DPSCs odontogenic differentiation mediated by TrkB

In our investigation of the combination of TNF α and CTX-B treatment compared to the control, we further aimed to identify the transcriptomic differences associated with the involvement of the BDNF-TrkB pathway related to inflammation. The gene heatmap and transcripts heatmap revealed distinct high and low gene expression patterns between the control and TNF α + CTX-B groups (**Figure 7A–B** [7A](#)–**7B**). The volcano plot of differentially expressed genes (DEGs) illustrated the significant downregulation and upregulation in gene expression between the control and TNF α + CTX-B groups (**Figure 7C–D** [7C](#)–**7D**).

We identified significantly enriched Gene Ontology (GO) terms in the categories of biological process, cellular component, and molecular function between these groups (**Figure 7E** [7E](#)). Notably, there was a distinct difference in membrane-related cellular functions and molecular functions related to protein binding. The GO enrichment pathway analysis revealed that the highest p-values were associated with the extracellular space and region (**Figure 8A** [8A](#)). Furthermore, Reactome pathway analysis indicated high p-values for pathways related to TNF signaling (**Figure 8B** [8B](#)).

Sashimi plots comparing control and TNF α + CTX-B treated dental pulp stem cells (DPSCs) highlighted differentially spliced exons in genomic regions of interest (**Figure 6D** [6D](#), E, F). The transcriptional response of TNF α + CTX-B treated odontoblast-like differentiated DPSCs showed significant downregulation in TCFs and ATF4 and upregulation in signal transducers and activators of STAT1 and STAT3 (**Figure 8C** [8C](#)). These findings suggest that TNF α and CTX-B treatment induces significant transcriptomic alterations, particularly affecting pathways involved in cellular signaling and extracellular matrix interactions.

Materials and methods

Chemicals and reagents

Human DPSCs were purchased from Lonza, Pharma & Biotech (Cat. # PT-5025). MEM- α , DMEM, PBS, fetal bovine serum, L-glutamine, and Antibiotic–Antimycotic were procured from Gibco™ Fisher Scientific (Waltham, MA, USA). Poly-D-Lysine coated (BioCoat™, 12 mm) round German glass coverslips were purchased from Corning™ Fisher Scientific (Cat. # 354087; Waltham, MA, USA). RIPA buffer was from Cell Signaling Technology (Danvers, MA, USA). Various antibodies were procured: anti-TrkA, anti-phospho-TrkA, anti-TrkB, anti-phospho-TrkB, anti-BDNF, and β -tubulin from BioLegend (San Diego, CA, USA), and anti- β -actin from Fisher Scientific (Waltham, MA, USA), anti-GFP and anti-STRO-1 from Santa Cruz (Dallas, Texas, USA). Fluorescent secondary antibodies were from Life Technologies (Grand Island, NY, USA). TNF α was from Invitrogen, Fisher

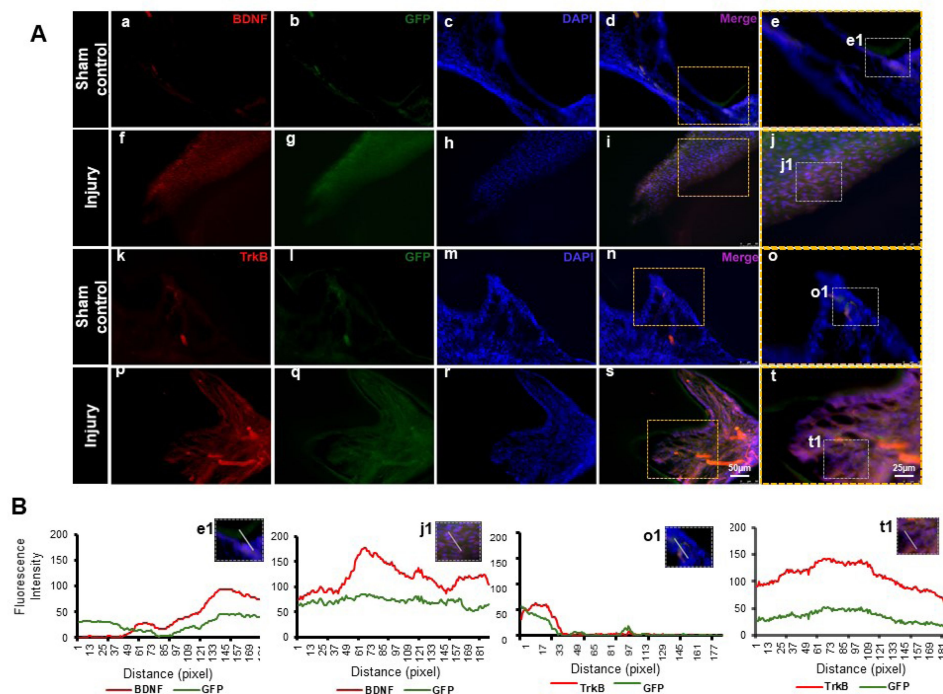


Figure 6.

Detection of BDNF and TrkB in GFP-tagged cells transplanted in the pulp-capping mouse model.

(A) Representative image of Sham control of the pulp-capping mouse model. (a-j) Sham control vs (k-t) representative image of the transplantation of CRISPR-engineered BDNF-overexpressing DPSCs in the pulp-capping mouse model. Scale bars: 25 and 50 μ m. (B) Line graphs showing the co-localization of BDNF and GFP [B(e1 and j1)] and of TrkB and GFP [B(o1 and t1)].

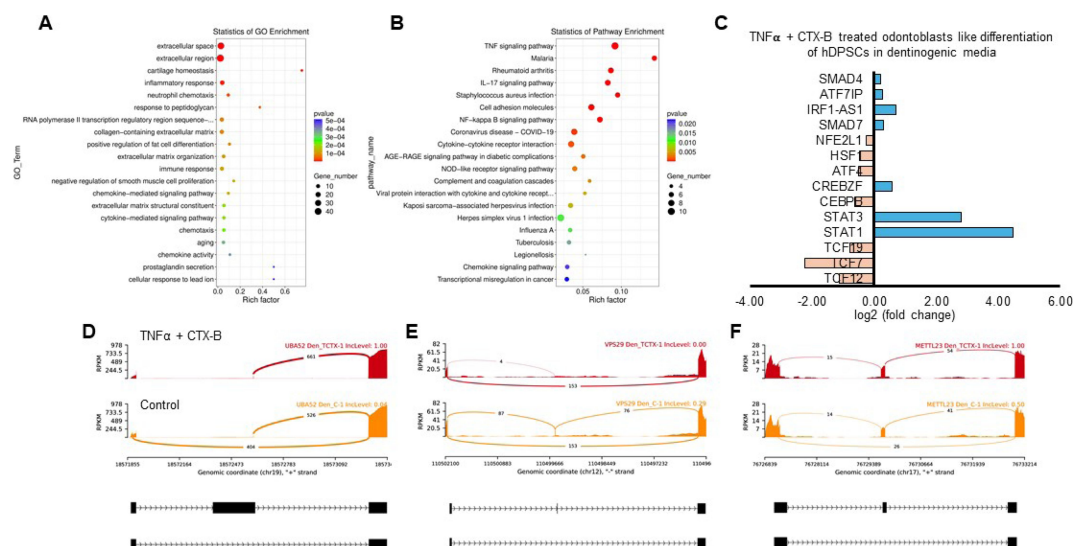


Figure 8.

Signaling pathways affected by TNF α -induced DPSCs mediated by TrkB inhibitor in odontoblasts like differentiation of DPSCs in dentinogenic media.

(A) A dot plot showing the GO terms enriched in the dataset. The size of the dots represents the number of DEGs, while the color indicates the p-value, with darker colors representing more significant enrichment depending on the rich factor, which is the ratio of the number of DEGs annotated in a GO term to the total number of genes annotated in that term. (B) the pathways enriched among the DEGs. The x-axis shows the rich factor, and the y-axis lists the pathways. The size of the dot indicates the number of genes involved in each pathway. (C) Next-generation RNA sequencing was done using poly-A-RNA sequencing technique. Histogram showing upregulated and activated transcription factors (blue) and repressed or down-regulated transcription factors (orange). Histogram showing the effect of CTX-B on the regulation of various transcription factors. Notably, TCF (7, 12, 19) was abolished with the treatment of CTX-B. (D-F) The Sashimi plots show the expression patterns of specific genes in response to TNF α +CTX-B treatment. Data were examined on sashimi plots, which revealed the number of variants and genomic mutations on chr19, chr12, and chr17 in TNF α +CTX-B treated cells in dentinogenic media against control. Red sashimi plots show variants in the TNF α +CTX-B treated group, and orange shows in the control. The lower black annotations are Read alignments of alternative isoforms and genomic regions of interest providing the gene model, with exons represented as thick blocks and introns as thin lines.

Scientific (Waltham, MA, USA), and a few other chemicals were from Fisher Chemical (Nazareth, PA, USA). BDNF CRISPR activation plasmid (h) (Cat# sc-400029-ACT) and Reagent System were purchased from Santa Cruz Biotechnology (Dallas, TX, USA).

Cell culture

All experiments were conducted with different sets of human dental pulp stem cells (DPSCs) 3–4 times, using cells at the 2nd and 3rd passages and cell proliferation was measured by counting the total number of cells. Commercially available human DPSCs isolated from third molars of adult donors (ages 29–30 years old) collected during the extraction of wisdom teeth, which were guaranteed through 10 population doublings, to express CD105, CD166, CD29, CD90, and CD73, and do not express CD34, CD45, and CD133 106–108; cultured in regular growth media (α MEM containing 10% FBS, 1% L-glutamine, and 1% antimycotic/antibiotic) at 37°C and 5% CO₂ for 3–4 days. After this initial period, the media was swapped to odontogenic media (DMEM containing 10% FBS, 1% L-glutamine and antimycotic/antibiotic, 50 μ g/mL ascorbic acid, 10 mM β -glycerophosphate, and 10 nM dexamethasone) at day 4 until day 17. TrkB antagonist (CTX-B, 200nM/ml) was treated every three days until day 17. Cells were treated with LPS, TNF α , LTA, C5a, and IL-5 at days 4 and 7. The CaMKII inhibitor (5 μ M/mL), or CaMKII protein (1 μ M/mL) were treated with dentinogenic media at days 4, 7, 10, and 14.

Quantitative real-time PCR analysis of odontogenic differentiation marker gene expression in dental pulp stem cells

DPSCs were cultured in a 6-well plate at 5×10^4 cells per well. The total mRNA was extracted using a RNeasy Mini Kit (74104, Qiagen), and the containing quantity of cDNA was measured using the NanoDrop 2000 (ND2000, Fisher Scientific). The Fast SYBR[™] Green Master Mix (4385616, ThermoFisher) was used to identify the cDNA sample according to the manufacture's protocol. Primer sequences (Integrated DNA Technologies) were used predesigned hGAPDH. (Forward: 5'-GGC ATC CAC TGT GGT CAT GAG-3', Reverse: 5'-TGC ACC ACC AAC TGC TTA GC-3'), hDSPP (Forward: 5'-CTG TTG GGA AGA GCC AAG ATA AG-3', Reverse: 5'-CCA AGA TCA TTC CAT GTT GTC CT-3'), and hDMP-1 (Forward: 5'-CAC TCA AGA TTC AGG TGG CAG-3', Reverse: 5'-TCT GAG ATG CGA GAC TTC CTA AA-3').

Immunocytochemistry

The seeded DPSCs were incubated in a 12-well plate with coverslips at 37°C in a CO₂ incubator until reaching 70–80% cell confluency. The coverslips were then fixed with 4% paraformaldehyde for 2 hours at 4°C. Blocking and permeabilization were performed using 5% normal goat serum and 0.01% Triton X in 0.01 M phosphate buffer solution (PBS) for 1 hour at room temperature. For primary antibody treatment, the specimens were treated overnight at 4°C with the following antibodies diluted in 5% serum: anti-TrkB (1:1000), anti-pTrkB (1:1000), and/or anti-STRO1 (1:1000). After the overnight incubation with primary antibodies, the secondary antibodies were applied for 2 hours with Alexa Fluor-594 anti-mouse IgG, Alexa Fluor-488 anti-rabbit IgG (1 μ g/mL), and/or DAPI (1 μ g/mL). The coverslips were mounted on glass slides, and images were taken using a Leica microscope. MATLAB (R2022a) software was used to measure fluorescence intensity and perform quantification.

BDNF CRISPR Activation Plasmid Transfection in hDPSCs

DPSCs were incubated in a 6-well plate with 3 ml of antibiotic-free standard growth medium at 37°C in a 5% CO₂ incubator until reaching 40–60% confluency in preparation for CRISPR activation plasmid transient transfection. For the preparation of transfection solutions, 2 μ g of plasmid DNA was diluted in Plasmid Transfection Medium (sc-108062) to a final volume of 150 μ l (Solution A) and mixed and incubated for 5 minutes at room temperature. In parallel, 10 μ l of UltraCruz[®] Transfection Reagent (sc-395739) was diluted in Plasmid Transfection Medium (sc-108062) to a

final volume of 150 μ l (Solution B) and incubated for 5 minutes at room temperature. Solutions A and B were then combined by vortexing immediately and incubated for 20 minutes at room temperature. For the transfection procedure, the cell culture medium was replaced with a fresh antibiotic-free growth medium, and 300 μ l of the Plasmid DNA/UltraCruz® Transfection Reagent Complex was added to each well. The cells were incubated for 48 hours under standard culture conditions. Protein analysis was conducted to confirm overexpression.

Western blot

To verify BDNF activation using CRISPR plasmid, cell lysates were prepared using RIPA buffer (50 mM Tris pH 7.6, 150 mM NaCl, 1% Triton X100, 1mM EDTA, 0.5% sodium deoxycholate, 0.1% SDS) containing protease inhibitor (Roche, Indianapolis, IN) and protein was measured using BCA method (Pierce™ BCA Protein Assay Kit, Thermo Fisher Scientific, Lenexa, KS). Equal amounts of protein (35 μ g) were loaded and separated by SDS-polyacrylamide gel electrophoresis and transferred onto nitrocellulose (Bio-Rad, CA). Blots were probed using polyclonal antibodies specific for BDNF or β -actin for overnight at 4°C and then washed and probed with secondary antibodies for 2 hours at RT. Odyssey CLx visualized the protein bands.

In cell Western assay

Human DPSCs were seeded in growth media at 15×10^3 cells/cm² in 96-well optical plates. At sub-confluency, cells were incubated in an antibiotic-free medium and treated as mentioned above. Then, cells were immediately fixed with 100% cold methanol (15 min) and saturated with 5% BSA (1.5 h). Cells were incubated overnight at 4 °C, anti-TrkA, anti-TrkB, anti-phospho-TrkA, anti-phospho-TrkB, or anti- β -tubulin. Cells were then washed (0.05% Tween-20/PBS) and incubated with respective IRDye-680RD or IRDye-800RD secondary antibody (1 h) at RT. After 5 washes, plates were dried and scanned at 700 and/or 800 nm (Odyssey CLx).

Animals

The Institutional Animal Care and Use Committee (IACUC) at the University of Illinois Chicago reviewed and approved all surgical and experimental procedures. C57BL/6 male mice, 6-8 weeks old (n = 12), were used for the experiments and purchased from Jackson Laboratory (#000461). The animals were housed in three per cage in a temperature-controlled room (23 $\pm$ 1°C, 12 h/12 h light/dark cycle) and maintained at the UIC animal facility. For the mouse molars, the mandibles of the C57BL/6 mice were collected for further experimentation (Protocol no.: 21-197).

Transplantation of CRISPR-engineered BDNF-overexpressing DPSC in the pulp-capping mouse model (In-Vivo)

C57BL/6 male mice aged 6 to 8 weeks were employed for this mouse model. Dentin of the lower left first molar was penetrated using a 0.3 mm rounded carbide burr drill operating at automatic speed. Subsequently, an 11 G needle was utilized to remove any remaining dentin upon exposure. The exposed dentin was then treated with BDNF CRISPR Activation Plasmid-transfected DPSCs embedded in collagen (from rat tail), followed by sealing with mineral trioxide aggregate (MTA) to shield the pulp from further inflammation. Animals were euthanized beginning 4 weeks post-injury, and their mandibles were excised. The molars were subsequently harvested for histological, molecular, and immunohistochemical analyses to evaluate the efficacy of the treatment in promoting dental pulp regeneration.

Micro-computed tomography (μ CT)

Mandibles from C57BL/6 male mice were collected from both the sham control group and the group receiving transplantation of CRISPR-engineered BDNF-overexpressing dental pulp stem cells (DPSCs) in the pulp-capping mouse model. Micro-CT (μ CT) analyses were conducted using the

commercial service at RUSH Hospital to assess the volume of regenerated dentin quantitatively. All animal phenotype analyses were blinded to mitigate potential examiner bias and ensure impartial evaluation.

Hematoxylin and eosin staining (H&E staining)

The mandibles of C57BL/6 mice were harvested and fixed in a 4% formaldehyde solution immediately after extraction. Subsequently, they were placed in decalcification solution within 15 mL Falcon tubes for two days, with daily changes in concentration (starting from 10% sucrose in PBS, progressing to 20% and then 30%). The mandibles were sectioned following decalcification, and histological staining protocols were optimized specifically for dental tissues. Sections underwent a sequence of staining procedures, including bleaching, eosin staining, and dehydration through graded ethanol solutions (ranging from 70% to 100% ethanol). Finally, the sections were cleared in xylene to prepare them for microscopic analysis and further histopathological evaluation.

Collection of human virgin teeth and caries teeth

Human wisdom tooth extractions were conducted by licensed dentists specializing in oral surgery. This study involving human specimens received ethical approval from the Endodontics clinic at the College of Dentistry, University of Illinois Chicago (Protocol no.: 2012-0588). Prior to participation, all patients provided informed written consent detailing the nature and objectives of the experimental procedures. Ethical considerations and patient confidentiality were strictly followed throughout the study, ensuring compliance with established guidelines and protocols for human research.

Immunohistochemistry

The extracted mandibles were embedded in an OCT cryostat sectioning medium and stored at -20°C for 2 hours before being cut into 20 μm thick sections. The mounted tissues were fixed with 4% paraformaldehyde for 1 hour at room temperature. Subsequently, blocking and permeabilization were performed using 10% normal goat serum (NGS) and 0.1% Triton X in 0.01 M phosphate buffer solution (PBS) for 1 hour at room temperature. For primary antibody incubation, the specimens were treated overnight at 4°C with mouse anti-CaMKII (1:200), rabbit anti-p-CaMKII (1:200), mouse anti-GFAP (1:200), and/or mouse anti-STRO1 (1:200), diluted in 10% NGS. Following overnight incubation with primary antibodies, the specimens were incubated for 2 hours at room temperature with secondary antibodies: Alexa Fluor-594 anti-mouse IgG, Alexa Fluor-488 anti-rabbit IgG (1 $\mu\text{g/mL}$), and/or DAPI (3 $\mu\text{g/mL}$). After secondary antibody treatment, the specimens were mounted on glass slides, and images were acquired using a Leica microscope. Fluorescence intensity was quantified using Image J software. Representative images were exported in TIFF format. Image J software and GraphPad Prism version 10 were employed for fluorescence intensity analysis.

RNA Sequencing (Poly-A RNA Seq)

RNA Sequencing was employed to comprehensively analyze the transcriptome, focusing specifically on mRNA, utilizing services provided by LC Biosciences (Houston, Texas). This method capitalizes on the polyadenylated (poly-A) tails present at the 3' ends of eukaryotic mRNA molecules. Dental pulp stem cells (DPSCs) underwent a 10-day differentiation protocol with or without specific treatments. Post-differentiation, total RNA was extracted from the odontogenic differentiated DPSCs for sample preparation. LC Biosciences conducted rigorous assessments to evaluate the extracted RNA's quantity and quality, followed by poly-A selection to enrich for mRNA transcripts. Subsequent cDNA synthesis facilitated library preparation, paving the way for high-throughput sequencing to explore the transcriptomic landscape in depth. This approach enabled the identification and characterization of gene expression patterns associated with the differentiation of DPSCs under various experimental conditions.

Statistical analysis

The statistical analyses were performed on at least 3 independent experiments with duplicates or triplicates, and statistical significance was determined using one-way analysis of variance (ANOVA) followed by post-hoc Dunnett's test (SAS 9.4) to compare the different treatments and their respective controls (p-value of 0.05 or less was considered statistically significant). In addition, the data were analyzed using Tukey's test to determine statistical significance between the groups. For quantification of immunofluorescence staining intensity, ImageJ 1.49v software was used. Fixed areas of 1 mm × 1 mm or 2 mm × 2 mm were selected to analyze differentiated cells' number or fluorescence intensity.

Discussion

Odontoblastic differentiation of DPSCs in response to caries injury takes place within an inflammatory environment (Chmielewsky et al., 2016 [↗](#)). However, there is still limited information on the role of inflammation in reparative dentinogenesis and the biology of DPSCs. Our findings demonstrate the activation of TrkB during the caries process. Consistent with this, our in vitro data confirmed that various inflammatory mediators increase TrkB expression and activation in DPSCs. These findings suggest that BDNF/TrkB directly affects the pulpal response to caries. In vivo, TrkB is more highly expressed in inflamed pulp compared to normal tissue. The transplantation of CRISPR-engineered BDNF-overexpressing DPSCs in a pulp-capping mouse model significantly enhances dentin regeneration and repair, as demonstrated by increased dentin volume and improved tissue morphology. Furthermore, RNA sequencing reveals significant transcriptomic changes in TNFα-treated DPSCs, affecting cellular signaling and extracellular matrix pathways. These results underscore the critical role of TrkB in inflammatory responses and dentin repair, suggesting potential for targeted TrkB modulation in dental tissue regeneration.

TNFα and LTA are known to mediate inflammatory responses and contribute to the recruitment and activation of immune cells at the injury site (Farges et al., 2015 [↗](#); Kawai & Akira, 2010 [↗](#)). This inflammatory response is crucial for eliminating pathogens and debris, setting the stage for tissue repair. In the context of dentin formation, these cytokines promote the differentiation and activity of odontoblast-like cells, which are responsible for producing the reparative dentin matrix. (Durand et al., 2006 [↗](#); Keller et al., 2010 [↗](#)) Lipopolysaccharides (LPS), components of the outer membrane of Gram-negative bacteria, also stimulate an inflammatory response when they infiltrate the dental pulp. (Brodzikowska et al., 2022 [↗](#)) The presence of LPS triggers the release of various pro-inflammatory cytokines, further enhancing the recruitment of immune cells and promoting the processes necessary for tertiary dentin formation. (Widbiller et al., 2018 [↗](#)) These pro-inflammatory cytokines coordinate a complex cascade of events that lead to the regeneration of dentin, highlighting their importance in dental tissue repair mechanisms.

Our study confirms the impact of pro-inflammatory stimuli on the expression and activation of TrkB in DPSCs. Immunofluorescence analysis (**Figure 1** [↗](#)) demonstrated that exposure to LPS, LTA, and TNFα significantly upregulated both TrkB and its phosphorylated form, p-TrkB, compared to the control. This suggests that these inflammatory agents can activate TrkB signaling pathways in DPSCs, potentially influencing their behavior and differentiation. TNFα showed the most pronounced effect, indicating its potent role in activating TrkB signaling. These findings are consistent with previous studies indicating that TNFα can enhance TrkB expression and activation in various cell types, promoting cellular responses to inflammation (Irfan et al., 2024 [↗](#); Kim, Irfan, Hossain, George, et al., 2023; Paula-Silva et al., 2009 [↗](#)). These results suggest that pro-inflammatory stimuli such as LPS, LTA, and TNFα can significantly enhance TrkB and p-TrkB expression in DPSCs.

Furthermore, our data demonstrate that significant upregulation of TrkA and TrkB expressions in dentinogenic media in response to inflammatory components. TrkB expression was notably increased by LPS, LTA, TNF α , and C5a, suggesting widespread receptor activation. This upregulation may play a critical role in mediating the cellular responses to inflammation, potentially influencing the behavior of DPSCs in the context of dental pulp inflammation and regeneration. Particularly, TNF α induced the highest TrkB upregulation in both media types, emphasizing the role of TrkB in mediating responses to neurotrophic factors and involvement in inflammatory pathways (Irfan, Kim, Druzinsky, et al., 2022). The differential expression in dentinogenic media indicates enhanced receptor sensitivity and signaling in an inflammatory environment (J. Zhang et al., 2016 [DOI](#)). A dose-dependent increase in TrkB expression was observed with escalating TNF α concentrations, with significant upregulation at 10 ng/mL, beyond which further increases in TNF α did not enhance expression. This suggests that higher doses are unnecessary to avoid inflammation related to degeneration or infection (Kim, Irfan, Hossain, George, et al., 2023). The time-dependent response to a fixed TNF α concentration showed a peak at day 7, indicating that a minimum of 7 days of differentiation is necessary for culturing DPSCs in inflammatory responses related to TrkB expression. In vivo, TrkB expression was significantly higher in the inflamed pulp tissue of carious teeth, indicating that TrkB may play a critical role in the response to dental caries and pulp inflammation. This suggests that BDNF-TrkB signaling could be crucial in the neurogenic and angiogenic processes involved in pulp tissue repair (Bar et al., 2021 [DOI](#); De Moraes et al., 2018 [DOI](#)).

The transplantation of CRISPR-engineered BDNF-overexpressing DPSCs has demonstrated promising results in enhancing dentin regeneration in a pulp-capping mouse model. Both micro-CT images and H&E staining analyses indicate that the overexpression of brain-derived neurotrophic factor (BDNF) via CRISPR engineering significantly promotes dentin repair at the injury site. BDNF is known for its role in molecular and physiological involvement (Lu et al., 2014 [DOI](#)). Its application in dental tissue engineering suggests that it may also play a crucial role in odontogenesis and dentinogenesis (Kim, Irfan, Hossain, George, et al., 2023). The significant increase in dentin volume in the micro-CT experiment observed in this study supports the hypothesis that BDNF can enhance the regenerative capabilities of DPSCs, leading to improved outcomes in dentin repair. Enhanced dentinogenesis was evidenced by the well-organized and structurally similar new dentin formation in the transplantation group compared to the native dentin structure. The histological analysis with H&E staining further verified these findings, showing a marked improvement in dentin formation and reduced inflammation in the BDNF-DPSC transplantation group compared to the control group. This suggests that BDNF enhances dentin regeneration, creating a conducive environment for tissue repair. Several potential mechanisms could explain that BDNF may enhance the proliferation and differentiation of DPSCs into odontoblast-like cells, promoting reparative dentin formation (Tsutsui, 2020 [DOI](#)). Moreover, the neurotrophic properties of BDNF could support the survival and function of sensory nerve fibers within the pulp, which may play a role in regulating the repair process (Bathina & Das, 2015 [DOI](#)).

In vitro experiments with DPSCs further supported these findings, demonstrating that TrkB expression is significantly upregulated in response to TNF α treatment. This suggests that inflammation enhances the neurogenic potential of DPSCs through TrkB signaling. The co-expression of TrkB in both carious teeth and DPSCs highlights the potential of targeting the BDNF-TrkB pathway in dental regenerative therapies. Modulating the inflammatory environment to enhance TrkB signaling could improve the regenerative properties of DPSCs, offering promising strategies for managing dental caries and pulpitis. The combined treatment of TNF α and CTX-B elucidates the significant transcriptional reprogramming compared with the control. The volcano plots highlight a substantial number of differentially expressed genes (DEGs), indicating both upregulation and downregulation of gene expression compared to control. The GO enrichment analysis reveals that these DEGs are predominantly associated with biological processes such as extracellular matrix organization, inflammatory response, and cytokine-mediated signaling

pathways. This is further supported by the GO term statistics, where notable terms like extracellular space and cartilage homeostasis are enriched, suggesting alterations in the cellular microenvironment and immune response.

Additionally, the pathway enrichment analysis identifies critical pathways such as TNF signaling, NF- κ B signaling, and cytokine-cytokine receptor interaction, underscoring the crucial role of inflammatory signaling in the observed transcriptional changes. Finally, the gene expression analysis under TNF α + CTX-B treatment provides specific examples of genes with altered expression patterns, confirming the extensive impact of the treatment at the molecular level. These findings suggest that TNF α treatment mediated by BDNF/TrkB pathway induces a complex network of transcriptional changes that modulate the extracellular matrix and inflammatory signaling pathways.

Conclusion

The findings of this study have significant clinical implications for dental regenerative therapies. The use of CRISPR-engineered BDNF-overexpressing DPSCs could offer a novel and effective approach for enhancing dentin repair and regeneration in cases of dental pulp injury. This approach could potentially reduce the need for more invasive treatments, such as root canal therapy, and improve outcomes for patients with dental trauma or disease. Further studies are necessary to investigate the long-term effects of BDNF overexpression on dentin regeneration and to explore the underlying mechanisms driving this enhanced repair process. Additionally, assessing the potential for clinical translation of this approach will be crucial in determining its viability as a therapeutic option for dental pulp injuries and related conditions.

Data Availability Statement

The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Additional information

Author Contributions

J.H.K., M.I., and S.C. contributed to the conception, design, data acquisition, analysis, and interpretation and drafted and critically revised the manuscript. J.H.K and M.I. conducted most of the experiments and contributed to data acquisition and analysis of the experiment, while S.K and A.P. contributed to partial experiments, contributed data analysis, and critically wrote and edited the manuscript. S.C. designed the original concept, contributed to data acquisition and interpretation, and financially supported the project.

Funding

This study was supported by the NIH/NIDCR Grant: R01 DE029816– SC.

References

- Bar J. K., Lis-Nawara A., Grelewski P. G (2021) **Dental Pulp Stem Cell-Derived Secretome and Its Regenerative Potential** *International Journal of Molecular Sciences* **22**:12018 <https://doi.org/10.3390/ijms222112018>
- Bathina S., Das U. N (2015) **Brain-derived neurotrophic factor and its clinical implications** *Archives of Medical Science: AMS* **11**:1164–1178 <https://doi.org/10.5114/aoms.2015.56342>
- Bowen W. H., Burne R. A., Wu H., Koo H (2018) **Oral Biofilms: Pathogens, Matrix, and Polymicrobial Interactions in Microenvironments** *Trends in Microbiology* **26**:229–242 <https://doi.org/10.1016/j.tim.2017.09.008>
- Brodzikowska A., Ciechanowska M., Kopka M., Stachura A., Włodarski P. K (2022) **Role of Lipopolysaccharide, Derived from Various Bacterial Species, in Pulpitis—A Systematic Review** *Biomolecules* **12**:138 <https://doi.org/10.3390/biom12010138>
- Cappoli N., Tabolacci E., Aceto P., Dello Russo C (2020) **The emerging role of the BDNF-TrkB signaling pathway in the modulation of pain perception** *Journal of Neuroimmunology* **349**:577406 <https://doi.org/10.1016/j.jneuroim.2020.577406>
- Chmielewski F., About I., Chung S.-H (2016) **Pulp Fibroblasts Control Nerve Regeneration through Complement Activation** *Journal of Dental Research* **95**:913–922 <https://doi.org/10.1177/0022034516643065>
- De Moraes J. K., Wagner V. P., Fonseca F. P., Vargas P. A., De Farias C. B., Roesler R., Martins M. D. (2018) **Uncovering the role of brain-derived neurotrophic factor/tyrosine kinase receptor B signaling in head and neck malignancies** *Journal of Oral Pathology & Medicine* **47**:221–227 <https://doi.org/10.1111/jop.12611>
- Durand S. H., Flacher V., Roméas A., Carrouel F., Colomb E., Vincent C., Magloire H., Couble M.-L., Bleicher F., Staquet M.-J., Lebecque S., Farges J.-C. (2006) **Lipoteichoic Acid Increases TLR and Functional Chemokine Expression while Reducing Dentin Formation in In Vitro Differentiated Human Odontoblasts** *The Journal of Immunology* **176**:2880–2887 <https://doi.org/10.4049/jimmunol.176.5.2880>
- Farges J.-C., Alliot-Licht B., Renard E., Ducret M., Gaudin A., Smith A. J., Cooper P. R (2015) **Dental Pulp Defence and Repair Mechanisms in Dental Caries** *Mediators of Inflammation* **2015**:1–16 <https://doi.org/10.1155/2015/230251>
- Galler K. M., Weber M., Korkmaz Y., Widbiller M., Feuerer M (2021) **Inflammatory Response Mechanisms of the Dentine-Pulp Complex and the Periapical Tissues** *International Journal of Molecular Sciences* **22**:1480 <https://doi.org/10.3390/ijms22031480>
- Hang P.-Z., Zhu H., Li P.-F., Liu J., Ge F.-Q., Zhao J., Du Z.-M (2021) **The Emerging Role of BDNF/TrkB Signaling in Cardiovascular Diseases** *Life* **11**:70 <https://doi.org/10.3390/life11010070>
- Irfan M., Chung S (2023) **C5L2 modulates BDNF production in human dental pulp stem cells via p38 α pathway** *Scientific Reports* **13**:74 <https://doi.org/10.1038/s41598-022-27320-6>

- Irfan M., Kim J. H., Druzinsky R. E., Ravindran S., Chung S (2022) **Complement C5aR/LPS-induced BDNF and NGF modulation in human dental pulp stem cells** *Scientific Reports* **12**:2042 <https://doi.org/10.1038/s41598-022-06110-0>
- Irfan M., Kim J.-H., Marzban H., Reed D. A., George A., Cooper L. F., Chung S (2022) **The role of complement C5a receptor in DPSC odontoblastic differentiation and in vivo reparative dentin formation** *International Journal of Oral Science* **14**:7 <https://doi.org/10.1038/s41368-022-00158-4>
- Irfan M., Marzban H., Chung S (2024) **C5L2 CRISPR KO enhances dental pulp stem cell-mediated dentinogenesis via TrkB under TNF α -induced inflammation** *Frontiers in Cell and Developmental Biology* **12**:1338419 <https://doi.org/10.3389/fcell.2024.1338419>
- Kawai T., Akira S (2010) **The role of pattern-recognition receptors in innate immunity: Update on Toll-like receptors** *Nature Immunology* **11**:373–384 <https://doi.org/10.1038/ni.1863>
- Keller J.-F., Carrouel F., Colomb E., Durand S. H., Baudouin C., Msika P., Bleicher F., Vincent C., Staquet M.-J., Farges J.-C (2010) **Toll-like receptor 2 activation by lipoteichoic acid induces differential production of pro-inflammatory cytokines in human odontoblasts, dental pulp fibroblasts and immature dendritic cells** *Immunobiology* **215**:53–59 <https://doi.org/10.1016/j.imbio.2009.01.009>
- Kim J.-H., Irfan M., Hossain M. A., George A., Chung S (2023) **BDNF/TrkB Is a Crucial Regulator in the Inflammation-Mediated Odontoblastic Differentiation of Dental Pulp Stem Cells** *Cells* **12**:1851 <https://doi.org/10.3390/cells12141851>
- Kim J.-H., Irfan M., Hossain M. A., Shin S., George A., Chung S (2023) **LPS-induced inflammation potentiates dental pulp stem cell odontogenic differentiation through C5aR and p38** *Connective Tissue Research* <https://doi.org/10.1080/03008207.2023.2218944>
- Lin L. M., Rosenberg P. A (2011) **Repair and regeneration in endodontics: Repair and regeneration in endodontics** *International Endodontic Journal* **44**:889–906 <https://doi.org/10.1111/j.1365-2591.2011.01915.x>
- Lu B., Nagappan G., Lu Y. (2014) **BDNF and Synaptic Plasticity, Cognitive Function, and Dysfunction. In G. R. Lewin & B. D. Carter (Eds.) Neurotrophic Factors** **220**:223–250 https://doi.org/10.1007/978-3-642-45106-5_9
- Miranda-Lourenço C., Ribeiro-Rodrigues L., Fonseca-Gomes J., Tanqueiro S. R., Belo R. F., Ferreira C. B., Rei N., Ferreira-Manso M., De Almeida-Borlido C., Costa-Coelho T., Freitas C. F., Zavalko S., Mouro F. M., Sebastião A. M., Xapelli S., Rodrigues T. M., Diógenes M. J. (2020) **Challenges of BDNF-based therapies: From common to rare diseases** *Pharmacological Research* **162**:105281 <https://doi.org/10.1016/j.phrs.2020.105281>
- Mosaddad S. A., Tahmasebi E., Yazdanian A., Rezvani M. B., Seifalian A., Yazdanian M., Tebyanian H (2019) **Oral microbial biofilms: An update** *European Journal of Clinical Microbiology & Infectious Diseases* **38**:2005–2019 <https://doi.org/10.1007/s10096-019-03641-9>
- Nakashima M., Iohara K., Murakami M (2013) **Dental pulp stem cells and regeneration** *Endodontic Topics* **28**:38–50 <https://doi.org/10.1111/etp.12027>
- Paula-Silva F. W. G., Ghosh A., Silva L. A. B., Kapila Y. L (2009) **TNF- α Promotes an Odontoblastic Phenotype in Dental Pulp Cells** *Journal of Dental Research* **88**:339–344 <https://doi.org/10.1177/0022034509334070>

Peiseler M., Kubes P 2018) **Macrophages play an essential role in trauma-induced sterile inflammation and tissue repair** *European Journal of Trauma and Emergency Surgery* **44**:335–349 <https://doi.org/10.1007/s00068-018-0956-1>

Peres M. A., Macpherson L. M. D., Weyant R. J., Daly B., Venturelli R., Mathur M. R., Listl S., Celeste R. K., Guarnizo-Herreño C. C., Kearns C., Benzan H., Allison P., Watt R. G 2019) **Oral diseases: A global public health challenge** *The Lancet* **394**:249–260 [https://doi.org/10.1016/S0140-6736\(19\)31146-8](https://doi.org/10.1016/S0140-6736(19)31146-8)

Potdar P. D., Jethmalani Y. D 2015) **Human dental pulp stem cells: Applications in future regenerative medicine** *World Journal of Stem Cells* **7**:839–851 <https://doi.org/10.4252/wjsc.v7.i5.839>

Tsutsui T 2020) **Dental Pulp Stem Cells: Advances to Applications. Stem Cells and Cloning: Advances and Applications** Volume **13**:33–42 <https://doi.org/10.2147/SCCAA.S166759>

Widbiller M., Eidt A., Wölflick M., Lindner S. R., Schweikl H., Hiller K.-A., Buchalla W., Galler K. M. 2018) **Interactive effects of LPS and dentine matrix proteins on human dental pulp stem cells** *International Endodontic Journal* **51**:877–888 <https://doi.org/10.1111/iej.12897>

Zhang F., Kang Z., Li W., Xiao Z., Zhou X 2012) **Roles of brain-derived neurotrophic factor/tropomyosin-related kinase B (BDNF/TrkB) signalling in Alzheimer's disease** *Journal of Clinical Neuroscience* **19**:946–949 <https://doi.org/10.1016/j.jocn.2011.12.022>

Zhang J., Yao W., Hashimoto K 2016) **Brain-derived Neurotrophic Factor (BDNF)-TrkB Signaling in Inflammation-related Depression and Potential Therapeutic Targets** *Current Neuropharmacology* **14**:721–731 <https://doi.org/10.2174/1570159X14666160119094646>

Author information

Ji Hyun Kim[#]

Department of Oral Biology, College of Dentistry, University of Illinois Chicago, Chicago, United States

[#]Co-first authors

Muhammad Irfan[#]

Department of Oral Biology, College of Dentistry, University of Illinois Chicago, Chicago, United States

[#]Co-first authors

Sreelekshmi Sreekumar

Department of Oral Biology, College of Dentistry, University of Illinois Chicago, Chicago, United States

Stephanie Kim

Department of Oral Biology, College of Dentistry, University of Illinois Chicago, Chicago, United States

Atsawasuwan Phimon

Department of Orthodontics, College of Dentistry, University of Illinois Chicago, Chicago, United States

Seung Chung

Department of Oral Biology, College of Dentistry, University of Illinois Chicago, Chicago, United States

ORCID iD: [0000-0003-1449-8762](https://orcid.org/0000-0003-1449-8762)

For correspondence: chungsh@uic.edu

Editors

Reviewing Editor

Moses Chao

New York University Langone Medical Center, New York, United States of America

Senior Editor

Dolores Shoback

University of California, San Francisco, San Francisco, United States of America

Reviewer #1 (Public review):

This work employs both in vitro and in vivo/transplant methods to investigate the contribution of BDNF/TrkB signaling to enhancing differentiation and dentin-repair capabilities of dental pulp stem cells in the context of exposure to a variety of inflammatory cytokines. A particular emphasis of the approach is the employment of dental pulp stem cells in which BDNF expression has been enhanced using CRISPR technology. Transplantation of such cells is said to improve dentin regeneration in a mouse model of tooth decay.

The study provides several interesting findings, including demonstrating that exposure to several cytokines/inflammatory agents increases the quantity of (activated) phospho-Trk B in dental pulp stem cells.

However, a variety of technical issues weaken support for the major conclusions offered by the authors. These technical issues include the following:

(1) It remains unclear exactly how the cytokines tested affect BDNF/TrkB signaling. For example, in Figure 1C, TNF-alpha increases TrkB and phospho-TrkB immunoreactivity to the same degree, suggesting that the cytokine promotes TrkB abundance without stimulating pathways that activate TrkB, whereas in Figure 2D, TNF-alpha has little effect on the abundance of TrkB, while increasing phospho-TrkB, suggesting that it affects TrkB activation and not TrkB abundance.

(2) I find the histological images in Figure 3 to be difficult to interpret. I would have imagined that DAPI nuclear stains would reveal the odontoblast layer, but this is not apparent. An adjacent section labeled with conventional histological stains would be helpful here. Others have described Stro-1 as a stem cell marker that is expressed on a minority of cells associated with vasculature in the dental pulp, but in the images in Figure 3, Stro-1 label is essentially co-distributed with DAPI, in both control and injured teeth, indicating that it is expressed in

nearly all cells. Although the authors state that the Stro-1-positive cells are associated with vasculature, but I see no evidence that is true.

(3) The data presented convincingly demonstrate that they have elevated BDNF expression in their dental pulp stem cells using a CRISPR-based approach I have a number of questions about these findings. Firstly, nowhere in the paper do they describe the nature of the CRISPR plasmid they are transiently transfecting. Some published methods delete segments of the BDNF 3'-UTR while others use an inactivated Cas9 to position an active transactivator to sequences in the BDNF promoter. If it is the latter approach, transient transfection will yield transient increases in BDNF expression. Also, as BDNF employs multiple promoters, it would be helpful to know which promoter sequence is targeted, and finally, knowing the identity of the guide RNAs would allow assessment for the potential of off-target effects I am guessing that the investigators employ a commercially obtained system from Santa Cruz, but nowhere is this mentioned. Please provide this information.

(4) Another question left unresolved is whether their approach elevated BDNF, proBDNF, or both. Their 28 kDa western blot band apparently represents proBDNF exclusively, with no mature BDNF apparent, yet only mature BDNF effectively activates TrkB receptors. On the other hand, proBDNF preferentially activates p75NTR receptors. The present paper never mentions p75NTR, which is a significant omission, since other investigators have demonstrated that p75NTR controls odontoblast differentiation.

(5) In any case, no evidence is presented to support the conclusion that the artificially elevated BDNF expression has any effect on the capability of the dental pulp stem cells to promote dentin regeneration. The results shown in Figures 4 and 5 compare dentin regeneration with BDNF-over-expressing stem cells with results lacking any stem cell transplantation. A suitable control is required to allow any conclusion about the benefit of over-expressing BDNF.

(6) Whether increased BDNF expression is beneficial or not, the evidence that the BDNF-overexpressing dental pulp stem cells promote dentin regeneration is somewhat weak. The data presented indicate that the cells increase dentin density by only 6%. The text and figure legend disagree on whether the p-value for this effect is 0.05 or 0.01. In either case, nowhere is the value of N for this statistic mentioned, leaving uncertainty about whether the effect is real.

(7) The final set of experiments applies transcriptomic analysis to address the mechanisms mediating function differences in dental pulp stem cell behavior. Unfortunately, while the Abstract indicates "we conducted transcriptomic profiling of TNF α -treated DPSCs, both with and without TrkB antagonist CTX-B" that does not describe the experiment described, which compared the transcriptome of control cells with cells simultaneously exposed to TNF- α and CTX-B. Since CTX-B blocks the functional response of cells to TNF- α , I don't understand how any useful interpretation can be attached to the data without controls for the effect of TNF alone and CTX-B alone.

<https://doi.org/10.7554/eLife.105153.1.sa2>

Reviewer #2 (Public review):

Summary:

In this manuscript, the authors investigate the potential for overexpressing BDNF in dental pulp stem cells to enhance dentin regeneration. They suggest that in the inflammatory environment of injured teeth, there is increased signaling of TrkB in response to elevated levels of inflammatory molecules.

Strengths:

The potential application to dentin regeneration is interesting.

Weaknesses:

There are a number of concerns with this manuscript to be addressed.

(1) Insufficient citation of the literature. There is a vast literature on BDNF-TrkB regulating survival, development, and function of neurons, yet there is only one citation (Zhang et al 2012) which is on Alzheimer's disease.

(2) There are several incorrect statements. For example, in the introduction (line 80) TrkA is not a BDNF receptor.

(3) Most important - Specific antibodies must be identified by their RRID numbers. To state that "Various antibodies were procured:... from BioLegend" is unacceptable, and calls into question the entire analysis. Specifically, their Western blot in Figure 4B indicates a band at 28 kDa that they say is BDNF, however the size of BDNF is 14 kDa, and the size of proBDNF is 32 and 37 kDa, therefore it is not clear what they are indicating at 28 kDa. The validation is critical to their analysis of BDNF-expressing cells.

(4) Figure 2 indicates increased expression of TrkB and TrkA, as well as their phosphorylated forms in response to inflammatory stimuli. Do these treatments elicit increased secretion of the ligands for these receptors, BDNF and NGF, respectively, to activate their phosphorylation? Or are they suggesting that the inflammatory molecules directly activate the Trk receptors? If so, further validation is necessary to demonstrate that.

(5) Figure 7 - RNA-Seq data, what is the rationale for treatment with TNF+ CTX-B? How does this identify any role for TrkB signaling? They never define their abbreviations, but if CTX-B refers to cholera toxin subunit B, which is what it usually refers to, then it is certainly not a TrkB antagonist.

<https://doi.org/10.7554/eLife.105153.1.sa1>

Reviewer #3 (Public review):

In general, although the authors interpret their results as pointing towards a possible role of BDNF in dentin regeneration, the results are over-interpreted due to the lack of proper controls and focus on TrkB expression, but not its isoforms in inflammatory processes. Surprisingly, the authors do not study the possible role of p75 in this process, which could be one of the mechanisms intervening under inflammatory conditions.

(1) The authors claim that there are two Trk receptors for BDNF, TrkA and TrkB. To date, I am unaware of any evidence that BDNF binds to TrkA to activate it. It is true that two receptors have been described in the literature, TrkB and p75 or NGFR, but the latter is not TrkA despite its name and capacity to bind NGF along with other neurotrophins. It is crucial for the authors to provide a reference stating that TrkA is a receptor for BDNF or, alternatively, to correct this paragraph.

(2) The authors discuss BDNF/TrkB in inflammation. Is there any possibility of p75 involvement in this process?

(3) The authors present immunofluorescence (IF) images against TrkB and pTrkB in the first figure. While they mention in the materials and methods section that these antibodies were generated for this study, there is no proof of their specificity. It should be noted that most commercial antibodies labeled as anti-TrkB recognize the extracellular domain of all TrkB isoforms. There are indications in the literature that pathological and excitotoxic conditions

change the expression levels of TrkB-FI and TrkB-T1. Therefore, it is necessary to demonstrate which isoform of TrkB the authors are showing as increased under their conditions. Similarly, it is essential to prove that the new anti-p-TrkB antibody is specific to this Trk receptor and, unlike other commercial antibodies, does not act as an anti-phospho-pan-Trk antibody.

(4) I believe this initial conclusion could be significantly strengthened, without opening up other interpretations of the results, by demonstrating the specificity of the antibodies via Western blot (WB), both in the presence and absence of BDNF and other neurotrophins, NGF, and NT-3. Additionally, using WB could help reinforce the quantification of fluorescence intensity presented by the authors in Figure 1. It's worth noting that the authors fixed the cells with 4% PFA for 2 hours, which can significantly increase cellular autofluorescence due to the extended fixation time, favoring PFA autofluorescence. They have not performed negative controls without primary antibodies to determine the level of autofluorescence and nonspecific background. Nor have they indicated optimizing the concentration of primary antibodies to find the optimal point where the signal is strong without a significant increase in background. The authors also do not mention using reference markers to normalize specific fluorescence or indicating that they normalized fluorescence intensity against a standard control, which can indeed be done using specific signal quantification techniques in immunocytochemistry with a slide graded in black-and-white intensity controls. From my experience, I recommend caution with interpretations from fluorescence quantification assays without considering the aforementioned controls.

(5) In Figure 2, the authors determine the expression levels of TrkA and TrkB using qPCR. Although they specify the primers used for GAPDH as a control in materials and methods, they do not indicate which primers they used to detect TrkA and TrkB transcripts, which is essential for determining which isoform of these receptors they are detecting under different stimulations. Similarly, I recommend following the MIQE guidelines (Minimum Information for Publication of Quantitative Real-Time PCR experiments), so they should indicate the amplification efficiency of their primers, the use of negative and positive controls to validate both the primer concentration used, and the reaction, the use of several stable reference genes, not just one.

(6) Moreover, the authors claim they are using the same amounts of cDNA for qPCRs since they have quantified the amounts using a Nanodrop. Given that dNTPs are used during cDNA synthesis, and high levels remain after cDNA synthesis from mRNA, it is not possible to accurately measure cDNA levels without first cleaning it from the residual dNTPs. Therefore, I recommend that the authors clarify this point to determine how they actually performed the qPCRs. I also recommend using two other reference genes like 18S and TATA Binding Protein alongside GAPDH, calculating the geometric mean of the three to correctly apply the $2^{-\Delta\Delta Ct}$ formula.

(7) Similarly, given that the newly generated antibodies have not been validated, I recommend introducing appropriate controls for the validation of in-cell Western assays.

(8) The authors' conclusion that TrkB levels are minimal (Figure 2E) raises questions about what they are actually detecting in the previous experiments might not be the TrkB-FI form. Therefore, it is essential to demonstrate beyond any doubt that both the antibodies used to detect TrkB and the primers used for qPCR are correct, and in the latter case, specify at which cycle (Ct) the basal detection of TrkB transcripts occurs. Treatment with TNF-alpha for 14 days could lead to increased cell proliferation or differentiation, potentially increasing overall TrkB transcript levels due to the number of cells in culture, not necessarily an increase in TrkB transcripts per cell.

(9) Overall, there are reasonable doubts about whether the authors are actually detecting TrkB in the first three images, as well as the phosphorylation levels and localization of this

receptor in the cells. For example, in Figure 3 A to J, it is not clear where TrkB is expressed, necessitating better resolution images and a magnified image to show in which cellular structure TrkB is expressed.

(10) In Figure 4, the authors indicate they have generated cells overexpressing BDNF after recombination using CRISPR technology. However, the WB they show in Figure 4B, performed under denaturing conditions, displays a band at approximately 28kDa. This WB is absolutely incorrect with all published data on BDNF detection via this technique. I believe the authors should demonstrate BDNF presence by showing a WB with appropriate controls and BDNF appearing at 14kDa to assume they are indeed detecting BDNF and that the cells are producing and secreting it. What antibodies have been used by the authors to detect BDNF? Have the authors validated it? There are some studies reporting the lack of specificity of certain commercial BDNF antibodies, therefore it is necessary to show that the authors are convincingly detecting BDNF.

(11) While the RNA sequencing data indicate changes in gene expression in cells treated with TNF α +CTX-B compared to control, the authors do not show a direct relationship between these genetic modifications with the rest of their manuscript's argument. I believe the results from these RNA sequencing assays should be put into the context of BDNF and TrkB, indicating which genes in this signaling pathway are or are not regulated, and their importance in this context.

<https://doi.org/10.7554/eLife.105153.1.sa0>